

Structured Semantic Components



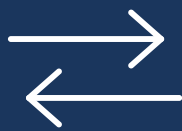
Event
Market phenomenon to be captured



Context
Condition under which the event is interpreted



Qualities
0-3 confirmation or filtering criteria



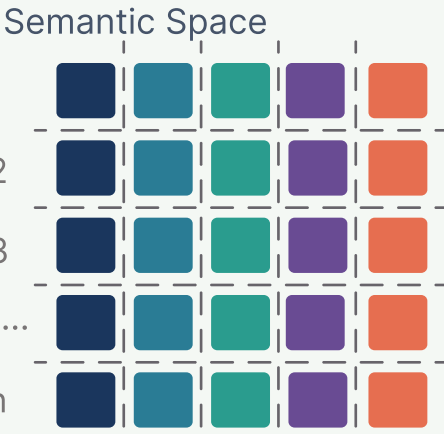
Direction
Expected relationship to future returns



Output
Tradable form of the resulting signal

$p = (e, c, Q, d, o)$

Sample Schema Plans(at each round)



randomly sample large amount of plans

Plan 1:

E1 + C3 + Q2 + D1 + O2

Plan 2:

E2 + C1 + Q4 + D2 + O3

Plan 3:

E5 + C2 + Q1 + D1 + O1

⋮

Plan N:

Ei + Cj + Qk + Dm + On

EXAMPLE CASE

Event Price Breakout

Context Recent High Region

Qualities Volume Confirmation + Multi-horizon Consistency

Direction Continuation

Output Cross-sectional Rank

Quota Scheduler

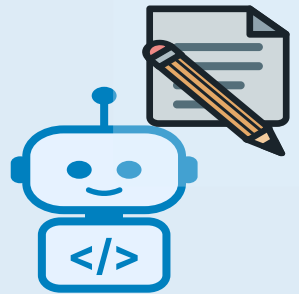
LightGBM Ensemble
Plan → Predicted Reward

Quota Selector
High Reward | Novelty | Local Mutation

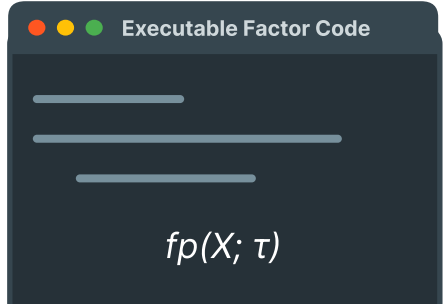
select batchsize plans to implement

Plan-to-Code Realization

LLM Agent:
Translate Factors



Executable Factor
Code



Multi-layer Validation & Backtest

- Contract Check
- Numerical Validity
- Leakage Check
- Backtest Reward

Objective Metrics

 IC ICIR

 RankIC RankICIR

Cumulative Reward Buffer and Re-training Pipeline

 **Scorer Model Re-training**

Retrain with Cumulative Dataset
Update Model Weights

Schema-Reward Memory Store

Schema p1	Reward r(p1)
Schema p2	Reward r(p2)
⋮	⋮
Schema pn	Reward r(pn)

$$D_t = D_{t-1} \cup B_t$$
$$B_t = \{(p_i, r(p_i)) \mid p_i \in \text{Batch}\}$$

Input: Validated Batch B(t)

New Evaluated (p, r(p))
Pairs